

Robotic Systems in Heritage Protection: An Anti-Fatigue Human-Robot Collaboration Exploration for Heritage Painting and Calligraphy Restoration

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ABSTRACT

The heritage restoration of paintings and calligraphy is a labour-intensive process, characterised by repetitive, low-intensity, and high-precision. Two core steps, “lift up the previous” and “fix and maintain”, can lead to muscle fatigue in the upper limbs. In this paper, we develop a human-in-the-loop based human-robot collaboration system that provides real-time force feedback to restorers based on their muscle load and motion intention, with the support strength changing to relieve fatigue. Different robot-aided positions on human arm are explored and a two-point elbow-supporting mechanism is evaluated as the optimal solution. Moreover, a novel muscle load calculation model is developed to integrate human factors into the HRC system with electromyography. The results of a usability experiment show the effectiveness of the proposed method in addressing the current anti-fatigue challenge in heritage paintings and calligraphy restoration.

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CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA
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ACM ISBN 979-8-4007-0129-0/23/10...\$15.00
<https://doi.org/10.1145/3584931.3606994>

CCS CONCEPTS

• **Human-centered computing** → *Interactive systems and tools*.

KEYWORDS

Human-Robot Collaboration; Heritage Restoration; Anti-Fatigue; Electromyography

ACM Reference Format:

Linyi Liu, Siqi Yang, Yixue Wang, Zhengtao Ma, Le Fang, Yonghao Long, and Stephen Jia Wang. 2023. Robotic Systems in Heritage Protection: An Anti-Fatigue Human-Robot Collaboration Exploration for Heritage Painting and Calligraphy Restoration. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3584931.3606994>

1 INTRODUCTION

Heritage restoration is an important way to protect cultural property. However, the current situation of heritage restoration is not optimistic [26]. According to the research, over 30 million cultural relics in China’s heritage system, half of them are in varying degrees of disrepair. Besides, only 2000 people in China are engaged in the restoration of heritage. It is particularly so in the paintings and calligraphy restoration. This work is a labour-intensive process, which is also characterised by repetitive, low-intensity and high-precision. Some researchers have used intelligent technologies in heritage protection [6, 13]. However, little has been done into restoring paintings and calligraphy.

There are two core steps “lift up the previous” and “fix and maintain”, which require the arm to hold a position for a prolonged period, leading to muscle fatigue and increased absolute error [12, 18]. Fatigue is a decrement in mental and/or physical performance caused by excessive muscle load and other factors [18]. Attributed to electromyography (EMG), quantified fatigue and real-time muscle load can be processed. For upper limb motion intention, EMG is one of the biosignals widely used for continuous prediction, reflecting intention around 50-100 ms before the actual movement occurs [5]. Different support positions exist for different movements and usage scenarios, such as supporting the lower arm while playing smartphone [23] and the wrist and elbow while writing [11].

Our team explores a human-robot collaboration (HRC) based solution, which combines the capabilities of both humans and robots to build a more efficient collaborative system with a common goal by incorporating the human-in-the-loop (HITL) structure [9, 15, 19]. This study proposed a HITL-based HRC system to give real-time force feedback to heritage painting and calligraphy restorer’s arm based on the level of muscle load and motion intention. In this paper, using EMG techniques, the support positions and support point numbers were experimentally explored, based on expert interviews and analysis of musculoskeletal models [24]. Combined with previous research, a muscle load calculation formula has been developed with low computability demand, which can quickly compute human factors into the HRC system [7]. On this basis, a HITL structure was built that could change the support strength to reduce muscle load, and the HRC system was evaluated to solve the current anti-fatigue challenge in heritage paintings and calligraphy restoration.

2 BACKGROUND

2.1 Core step of heritage painting and calligraphy restoration

Ancient Chinese painting and calligraphy contain four parts: backing, lining paper, painting, and mounting paper [22]. There are five main steps in its restoration process, including preparation, lifting up the previous, fixing and maintaining, colour restoration, and mounting [25]. This study focuses on two steps, “lift up the previous” and “fix and maintain”, which mainly involves repairing the lining paper. To accomplish this, restorers first lay a large piece of feather

paper on the surface of the painting and then flip the entire painting over. Subsequently, they use tools to remove the previous backing and lining paper, replacing it with a fresh layer. These two steps are not only the most important in painting and calligraphy restoration, but also the longest and require significant patience, precision, and slow speed. However, the repetitive operation of suspending the arm for extended periods inevitably leads to muscle fatigue.

Due to the complex damage situation and the near-unacceptable occurrence of mistakes, this work heavily relies on restorers’ expertise and experience for precise and meticulous restoration [5].

2.2 Influence of fatigue

Excessive muscle load may lead to decreased mental and physical performance, resulting in psychological and physical fatigue [18]. Such fatigue can cause decreased performance and sensitivity as well as increased error [12]. The operator’s ability to compensate for fatigue in the initial phase [17] is less effective as working hours increase [8].

Our team found similar results in expert interviews with painting and calligraphy restorers. Quoted from one restorer, “As the painting canvas is very thin, the restoration needs to be done carefully in case there is irreversible damage to the heritage, and muscle soreness is common because of the eight hours of continuous work each day.” Therefore, it is necessary to intervene to relieve upper limb fatigue. Robots have the ability to analyze human physiological signals, assess muscle load, and understand motion intention [3, 5, 18]. This inspires us to have robots working alongside human beings in restoration tasks because robots can accurately comprehend the situation and assist humans.

3 METHODS

3.1 HITL-based HRC system development

Our HITL-based HRC system provides real-time force feedback to restorers’ arms based on muscle load and motion intention. This enables efficient human-robot cooperation in repetitive and high-precision tasks. The human, acting as a crucial element in the loop, transmits tension, arm position, and muscle EMG signals to the robot. The robot processes this input, extracting muscle load, horizontal motion, and vertical intention (shown in Fig. 1). With this information, the robot gains a comprehensive understanding of

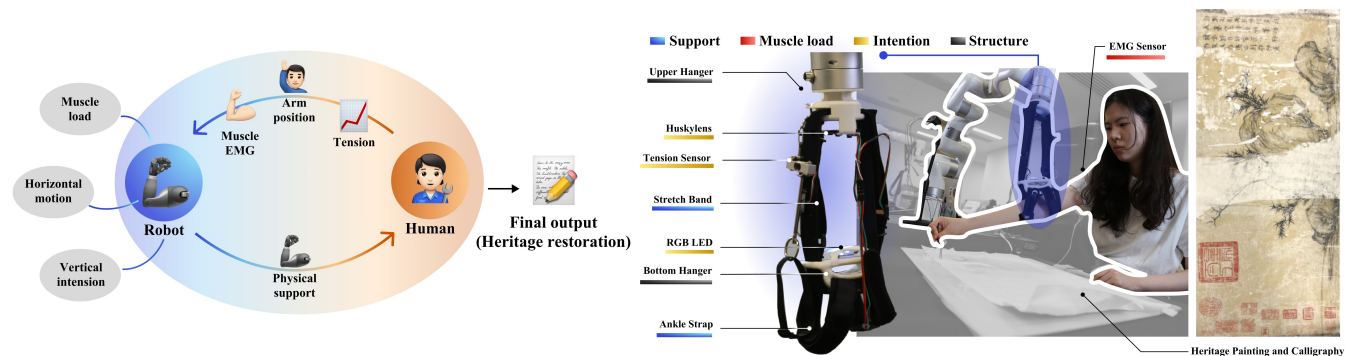


Figure 1: System concept and physical application

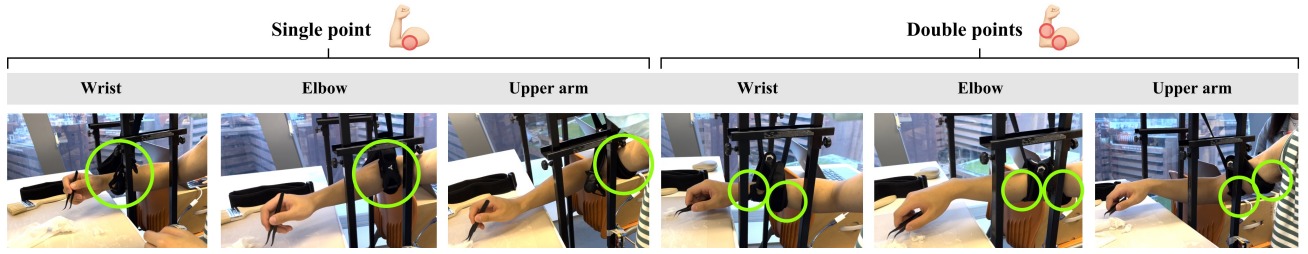


Figure 2: Experiment to explore the best position and number of support points

the restorer's state and specific requirements. By providing adaptive physical support, the robot can assist the restorer in performing tasks with greater ease, accuracy, and efficiency.

The HRC system consists of two main components: interactive computation logic and physical application. The system collects data from EMG and tension sensors, and processes it through the micro-controller, which sends commands to the robotic arm. And the system controls the movement of the robotic arm to give physical support for human arm. On the right side of Fig. 1, the physical application of system consists of hard and soft structures, sensors, and processors. Specifically, the hardware component includes an upper and bottom hanger connected by a stretch band, a tension sensor in the middle, an elbow strap for two points of support, and an EMG sensor attached to user's upper arm.

3.2 Support structure exploration

At the beginning, an experiment was conducted to determine the best position and number of support points. Fig. 2 illustrates the three support positions (wrist, elbow, and upper arm) and two numbers of support points (one-point and two-point). The subjects performed simulated restoration work for 10 minutes under six conditions. They were instructed to keep the testing arm stationary and perform horizontal moderate amplitude movements. During the experiment, the EMG data of the deltoid muscle was measured and presented in waveform graphs. Compare the EMG waveform graphs to assess the anti-fatigue effects and the interference of forearm movements on EMG measurements under different support types. In addition, the subjective comfort level under different support types was recorded using the Likert scale [14]. Based on the results, we found that the elbow was the best support location, and providing two support points was the most effective. Therefore, we integrated this support structure into the HRC system.

3.3 Muscle load estimation

Muscle fatigue causes an increase in lactic acid concentration, leading to a decrease in muscle fibre conduction velocity. These biochemical and physiological changes affect the properties of the EMG signal [10]. Our system has two main goals: real-time detection in a low-computing embedded environment, and catering to low-intensity, long-duration repetitive arm motions. Previous studies have explored various methods for muscle load detection, with high accuracy rates reported [1, 2]. However, the changes in EMG signal parameters at low load levels are ambiguous due to potential influences from factors other than muscle fatigue [21].

In our case, short-term muscle fatigue does not exhibit significant changes, but we observe a noticeable pattern in the amplitude of the EMG signal during the fatigue process. This amplitude change reflects increased muscle load due to arm suspension [16, 20].

Therefore, we proposed a novel method, which is to estimate real-time muscle load through catching amplitude change, rather than estimating muscle fatigue. Two classic indicators of EMG amplitude are adopted: mean absolute value (MAV), and root mean square (RMS) value. MAV is the mean value of the rectified EMG signal, indicating the overall state of amplitude. RMS represents the total power of the signal and is proportional to the overall muscle contraction [4]. The RMS and MAV values increase with increasing muscle force [1], but their relationship with muscle load needs to be examined. We tried different combination functions using these two indicators, and found the product results were most consistent with real-time muscle load perception. They are defined by following equations:

$$MAV = \frac{1}{N} \sum_{i=1}^N |x_i| \quad RMS = \sqrt{\frac{1}{N} \sum_{i=1}^N |x_i|^2} \quad (1)$$

$$Muscle \ Load = MAV \times RMS \quad (2)$$

In both equations x_i is the i^{th} sample of the signal in one window, and N is the size of the window.

From the expert interview, we conducted with a rehabilitation doctor, measuring the anterior and lateral bundles in the deltoid area can get effective results in our scenario. As shown in Fig. 3, We put three electrode pads on the bone, muscle top and muscle end of the deltoid muscle group in order.

3.4 Spatial support adaptation

The support adaptation is to detect muscle load in real-time during fatigue and give immediate support feedback in vertical motions. As mentioned in the prior section, the apparent amplitude change reflected the increase of muscle load at the time, which is also consistent with the insights obtained from the expert interview. What's more, "A suitable level of support can instantly change the high muscle load to low muscle load", noted by expert interviewees.

Support mechanism (shown in Fig. 3) can be divided into two parts: upward feedback and downward feedback. When the muscle load rises and exceeds the threshold, which means fatigue is going to appear, the robotic arm will move up to give more support strength until the muscle load falls. The threshold value is determined experimentally, overall it varies from 5 to 10 based on users'

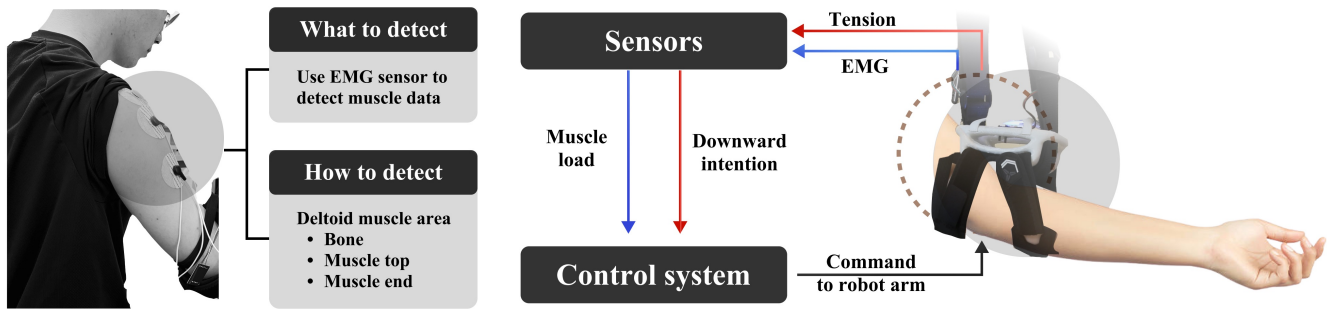


Figure 3: Muscle load detection method and mechanism of our system

initial status and personal characteristics. The downward feedback is designed to prevent over-support, in which case the human arm feels uncomfortable and has downward intention, resulting in an increased muscle load too. It is worth noting that muscle load change itself can not represent direction. So we add a tension sensor as an aid to monitor users' downward intention. Once increased muscle load and tension are detected, the robotic arm will know it is over-support and go down. Besides, it also gives users a certain degree of freedom in the vertical direction by considering downward motion intention.

To enhance the flexibility at the working surface, it is necessary to enable the robot arm to track the horizontal motion of the human arm. Using an AI camera on the upper hanger to recognize the position of an RGB LED set on the bottom hanger, the robotic arm can move following the horizontal movement of the user's arm.

4 EVALUATION

A usability test was conducted to verify whether the HRC system could relieve fatigue for users in heritage restoration work. This experiment was conducted by separating two sheets of paper glued together with tweezers to simulate a realistic scenario of restoring paintings and calligraphy. And comparing the amount of muscle load change before and after the experiment with and without the assistance of the HRC system.

4.1 Subjects

10 subjects (5 women and 5 men) participated in this experiment, and ranged in age from 23 to 26 years old, with an average age of 24.5 years old. All participants were from the Hong Kong Polytechnic University, had healthy physical conditions, no history of hand diseases, and were right-handed.

4.2 Experimental procedure

This was a single-factor experiment with two groups: an experimental group and a control group. The experimental group simulated repair work with the assistance of HRC system, while the control group performed the same tasks without assistance. Each subject participated in both groups randomly, with a full rest period between them.

Prior to the experiment, subjects rested for 3 minutes to stabilize their heart rate and reduce fatigue. The HRC system was applied in the first group, and after 5 minutes of practice time, the formal

experiment began. The initial muscle load value was measured, emphasizing average values from 10-second windows to minimize disturbances. According to the conclusion above, the initial value had to be less than 5 to proceed to the practice stage. If the initial value was less than 5, the subject proceeded to practice. Otherwise, the subject rested until the threshold was met.

Muscle load values were measured again after 20 minutes of task repetition. The Borg RPE Scale and interview questions were completed after the first group of tasks. The second set of experiments then began, repeating the same steps. The experiment took approximately 70 minutes in total.

4.3 Data processing and analysis

In this experiment, both the experimental group and the control group collected 10 sets of valid data. The initial muscle load value and Borg RPE scale were recorded for each subject, as well as their values after simulating restoration work for 20 minutes, and their feedback from interviews.

For each group's muscle load data, the increase was calculated by subtracting the initial value from the value at 20 minutes. Then, the growth amount of the control group was subtracted from that of the experimental group to calculate the difference. Positive differences indicated that the HRC system was effective in reducing muscle load growth. Finally, the difference percentage was calculated to reflect the objective performance of the system. We adopted the same method to analyze the Borg RPE scale data, to reflect the subjective performance of the system in relieving fatigue.

4.4 Result

The muscle load values of the experimental group and the control group increased by 4.75 and 8.64, respectively. After adding the HRC system, the muscle load growth was reduced by 3.89, with a reduction rate of 45.00%. The Borg RPE values for the experimental and control groups increased by 2.85 and 5.05, respectively, and with the assistance of the HRC system, the Borg RPE values decreased by 2.20, resulting in a decrease rate of 43.56%. Therefore, it can be seen that the HRC system indeed effectively relieved the fatigue in both objective and subjective aspects.

Three main findings emerged from the compilation of the interview results. First, most subjects felt that the assistance of the HRC system allowed them to reduce arm shake and focus more on hand

movements, thereby reducing fatigue and improving accuracy. Second, subjectively, time passed faster with assistance and felt longer without it. Third, although they feel distrustful and unskilled when using the system at the beginning, they can work well with the robot arm after a few minutes.

5 CONCLUSION AND DISCUSSION

This study developed a HITL-based HRC system for heritage painting and calligraphy restoration, providing a feasible HRC solution to address the challenge of anti-fatigue in this field. The system uses the optimal support method of two-point elbow support, and is equipped with a muscle load calculation formula suitable for low-computing power and embedded environments. The final results showed that the system objectively reduced the increase of muscle fatigue by 45.00% and subjectively reduced the increase of fatigue sensation by 43.56%. Our experiment has two limitations: Firstly, the muscle load threshold varies with individuals, and we are exploring personalized thresholds for different user groups. Secondly, the subjects lack expertise, necessitating experienced restorers or sufficient pre-practice.

ACKNOWLEDGMENTS

This research is funded by the PolyU Research Centre for Future (Caring) Mobility (P0042701); the University's Strategic Importance Project (Project Code: P0036851); and School of Design Collaborative Research Funding (2021-2023) (Project Code: P0035075), HKPolyU. This study is also supported by the Laboratory for Artificial Intelligence in Design (Project Code: RP2-5) under the InnoHK Research Clusters, Hong Kong Special Administrative Region Government. Thanks to editor, reviewers, and participants.

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